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Building mindfulness bottom-up: Meditation in natural settings supports open monitoring and attention restoration

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ABSTRACT

Mindfulness courses conventionally use effortful, focused meditation to train attention. In contrast, natural settings can effortlessly support state mindfulness and restore depleted attention resources, which could facilitate meditation. We performed two studies that compared conventional training with restoration skills training (ReST) that taught low-effort open monitoring meditation in a garden over five weeks. Assessments before and after meditation on multiple occasions showed that ReST meditation increasingly enhanced attention performance. Conventional meditation enhanced attention initially but increasingly incurred effort, reflected in performance decrements toward the course end. With both courses, attentional improvements generalized in the first weeks of training. Against established accounts, the generalized improvements thus occurred before any effort was incurred by the conventional exercises. We propose that restoration rather than attention training can account for early attentional improvements with meditation. ReST holds promise as an undemanding introduction to mindfulness and as a method to enhance restoration in nature contacts.

1. Introduction

Many people in modern societies routinely challenge their cognitive and emotional capabilities in efforts to meet the demands of their work and personal lives. These efforts draw down adaptive resources, like the ability to direct attention despite intrusive thoughts or external distractions. If the resources remain depleted, focus and performance will deteriorate, fatigue and chronic stress ensue, and health and well-being suffer (Cohen, 1980; Kaplan, 1995; von Lindern, Lymeus, & Hartig, 2017). Different approaches have been proposed to prevent these negative effects. Some aim to enable periodic relief from demands and promote restoration of depleted resources, like restorative environments approaches (e.g., Hartig, Mitchell, De Vries, & Frumkin, 2014; Kaplan, 1995). Others target a presumed need for individual training to strengthen the capabilities needed to meet demands, like mindfulness training (e.g., Brown, Ryan, & Creswell, 2007; Tang, Holzel, & Posner, 2015; cf. Kaplan, 2001; Tang & Posner, 2009). In this paper we offer an integration of these approaches that takes advantage of their respective strengths to overcome their respective weaknesses. In the following, we first consider the restorative environments and mindfulness training approaches alone, and then we indicate points of connection between the two. We draw on these connections in our integrated approach which we call restoration skills training (ReST).

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1.1. Restorative environments

Restoration refers to the replenishment of an adaptive resource that has been taxed in efforts to meet demands, so that the associated functional capabilities are reinstated (Hartig, 2017; Hartig et al., 2014; Kaplan, 1995). Given a need for restoration, some environments support its fulfillment better than others. Restorative environments are places that not only permit but facilitate restoration by supporting (1) a sense of psychological distance from stressors, distractions and demanding routines, and (2) pleasantly interesting experiences that engage attention effortlessly and evoke positive emotions (Hartig et al., 2011; Kaplan, 1995; Ulrich, 1983). In their attention restoration theory (ART), Kaplan and Kaplan (1989; Kaplan, 1995) termed the key restorative processes *being away* and *soft fascination*. Working together, these processes help to relieve the overworked adaptive systems, shield the experiential field from unwanted intrusions, stimulate soft and unconstrained attentiveness in the present, and aid psychophysiological stress recovery, all without imposing any additional self-regulation needs. Importantly, restorative experience is not just a matter of environmental qualities but of qualities of the transaction between individual and environment (Hartig, Korpela, Evans, & Gärling, 1997). The way someone engages with the environment is presumably amenable to training in relevant skills (Kaplan, 2001).

Most research on restorative environments has considered experiences with widely available natural settings, as with strolling in an urban park, visiting the countryside, or gardening (Hartig & Kahn, 2016; von Lindern et al., 2017). Research generally affirms that, for stressed and fatigued individuals, engaging with nature restores capabilities for directing attention and regulating emotions better than spending time in other commonly accessible environments (for reviews, see Bowler, Buyung-Ali, Knight, & Pullin, 2010; Hartig et al., 2014; Ohly et al., 2016; cf. Hartig & Jahncke, 2017). Many people can occasionally leave obligations aside, enter a restorative environment, and enjoy its benefits without any special skills or effort; however, the social, economic, geographic, architectural, and technological fabric of modern life tends to constrain opportunities for and the quality of restorative experiences (Hartig & Kahn, 2016; Staats, Jahncke, Herzog, & Hartig, 2016; von Lindern, 2017). Consequently, some studies have begun to consider ways to enhance engagement in limited contacts with nature (e.g., Duvall, 2011; Korpela, Savonen, Anttila, Pasanen, & Ratcliffe, 2017). Training in mindfulness skills could be one way to accomplish that.

1.2. Effortful mindfulness training

Mindfulness means a curious, yet detached, quality of attention to present experience (e.g., Bishop et al., 2004); that is, to connect with openness and acceptance with experiences while also maintaining some psychological distance in the experiential process. The psychological distance allows for breadth and flexibility of awareness and behavioral options. Similar to restorative experience, mindfulness can be considered as a state that can vary momentarily as people go through different activities and environments (Brown et al., 2007; Davidson & Kaszniak, 2015; Kaplan, 2001; Lutz, Jha, Dunne, & Saron, 2015; Tang & Posner, 2009). However, mindfulness is more commonly considered as a trait; a set of cognitive-behavioral skills or a neurocognitive capability that can be enhanced over weeks, months, or years with regular meditation exercise. As such, mindfulness training (like other training approaches) requires a significant initial investment of time, effort, and cognitive resources to establish practice habits and learn skills (Kaplan, 2001; Lutz et al., 2015; Lymeus, Lundgren, & Hartig, 2017; Malinowski, 2013; Tang & Posner, 2009). Once learned, however, mindfulness skills can help people to balance their resources and function more efficiently in many different situations. Mindfulness training can thus have a wide range of benefits (Eberth & Sedlmeier, 2012). Our focus here, however, is on attentional processes and outcomes, as several authors have proposed that attentional enhancements are fundamental to many of the other self-regulation and health enhancements that have been reported with meditation training (see e.g., Chiesa, Calati, & Serretti, 2011; Malinowski, 2013; Tang et al., 2015).

Many contemporary mindfulness courses for beginners emphasize so-called focused attention practice in which meditators try to sustain attention to given target stimuli, such as sensations with the breath, and repeatedly redirect attention when they lose their focus (Lutz, Slagter, Dunne, & Davidson, 2008; Lutz et al., 2015; Malinowski, 2013; Tang et al., 2015). During the exercises, experiences of fatigue and distraction are common (Frewen, Hargraves, DePierro, D'Andrea, & Flodrowski, 2016; Hasenkamp, Wilson-Mendenhall, Duncan, & Barsalou, 2012; Lutz et al., 2015). In fact, the experiences of fatigue and distraction are considered as opportunities for training in managing the intensity and direction of focus. With exercise on all or most days over several course weeks, focused attention practice is thought to train cognitive-behavioral attention control skills (e.g., Bishop et al., 2004; Kaplan, 2001) and stimulate gradual development of brain networks involved in cognitive control (Fox et al., 2016; Holzel et al., 2011; Lutz et al., 2008). Posner, Tang and colleagues (e.g., Posner, Rothbart, Rueda, & Tang, 2010; Tang & Posner, 2009) have termed this type of training "attention network training".

Other approaches to cognitive training have recently been criticized for conceptual shortcomings and over-extended promises (Rabipour & Raz, 2012; Simons et al., 2016), but the meditation literature has continued to accumulate evidence of enhanced performance on selective and executive attention tasks after weeks or months with regular exercise (Chiesa et al., 2011; Eberth & Sedlmeier, 2012; Gallant, 2016; Sedlmeier et al., 2012; though also see Lao, Kissane, & Meadows, 2016). Interestingly, some studies have reported such improvements after only one (Friese, Messner, & Schaffner, 2012), or as few as four (Zeidan, Johnson, Diamond, David, & Goolkasian, 2010), or five (Tang et al., 2007) meditation sessions.

Although many researchers construe the longer-term enhancements as resulting from attention network training, few have considered that the training can incur a cost. A recent study by Lymeus et al. (2017) found that many individuals who have weak attention to begin with struggle with the exercise requirements of the much researched Mindfulness-Based Stress Reduction (MBSR; Kabat-Zinn, 1990) course. Indeed, it makes sense that people who already exhaust themselves in trying to maintain focus and composure in their daily lives could be unwilling or unable to comply with requirements to spend any free time doing much the same

in focused attention exercises even with the prospect of attaining benefits in the long-term. Their primary need may be for restoration before starting an attention network training regimen.

1.3. Effortless mindfulness in nature

While focused attention practice constitutes the bulk of most mindfulness courses for beginners, many exercises also entail elements of a different type of practice, open monitoring, in which meditators use minimal effort to mindfully observe the changing stream of ongoing experience and attentional shifts from moment to moment (Lutz et al., 2008; Tang et al., 2015). Like focused attention practice, open monitoring requires a decision to meditate and maintenance of that decision over time in practice. In contrast to focused attention practice, however, open monitoring is presumably less effortful because meditators exercise little cognitive control (Fox et al., 2016; Lutz et al., 2015). Although open monitoring practice is not thought to train attention per se, it presumably develops the capability for curious yet detached connection with experience. Those processes can functionally support attention by enhancing contact with present experience and by facilitating disengagement from habitual and reactive thoughts, emotions, and behaviors that would distract and diminish connection with present experience. Posner, Tang and colleagues (e.g., Posner et al., 2010; Tang & Posner, 2009) have termed this type of training "attention state training" (in contrast to attention network training). Building on work by Kaplan (2001), they indicate links between attention state training in open monitoring-type meditation and restorative experiences in nature.

Theory in the mindfulness field and the restorative environments field converge in multiple ways. Both describe a particular quality of attention to present experience (Kaplan, 2001; Tang & Posner, 2009; cf. Bishop et al., 2004; Kaplan, 1995; Kaplan & Kaplan, 1989); that is, attention characterized by curiosity and soft fascination, respectively. Both also describe a way to manage stressors and distractions by obtaining some psychological distance from them – detachment and being away, respectively – rather than immediately attempting to address and eliminate them. While these transactional qualities apparently overlap between mindfulness and nature experience, the approaches differ in how they see the connection with present experience and the management of distraction being supported by top-down and bottom-up processes. Mindfulness research centers on the volitional practice of certain individual skills while restorative environments research focuses on processes that tend to appear spontaneously in certain types of environments. We propose that those top-down and bottom-up processes can converge to enhance each other, in effect supporting state mindfulness as well as restorative processes.

In an initial test of these ideas, Lymeus et al. (2017) had stressed students without previous meditation experience come in after a day of work to practice unguided mindfulness meditation for 15 min with or without simultaneously viewing images of natural scenery. The students did this every other week over a period of eight weeks, during which they also participated in a MBSR course, so their mindfulness skills were expected to improve gradually over the weeks. Control participants came and viewed the nature images with instructions to just rest, and they did not attend any mindfulness training. On average, participants who just rested with nature images showed improved attentional performance (assessed with a substitution test) after the sessions compared to before. The meditating participants showed no average improvement. This result indicates that mindfulness exercise for beginners can incur attentional effort and hamper any restoration otherwise expected with the 15-min rest session. The participants who practiced mindfulness without nature stimuli had increasingly negative effects of the sessions across the eight weeks. This result suggests that they exerted more effort in the meditation as they became more skilled. The participants who practiced mindfulness with nature images instead had increasingly positive effects of the sessions across the eight weeks. Meditation with natural stimuli can thus, given some basic mindfulness skill, offset some of the effort incurred by the exercises.

1.4. The present studies: context and purposes

Natural settings would not likely support beginners in the practice of focused attention meditation because a rich and softly fascinating environment would draw attention away from any given object of practice. The attentional demands of focused attention practice would also likely interfere with any restorative processes stimulated by the natural environment. Open monitoring practice, in contrast, would allow for attention to wander to different experiences in the environment. With open monitoring practice, we reasoned that natural stimuli can softly and effortlessly hold attention to various aspects of the present reality and stimulate mindful curiosity; and that a natural setting that supports a sense of being away from stressors, demands, and routine mental contents can facilitate mindful detachment. Thus, nature should support beginning meditators in open monitoring practice. We further reasoned that skills in curiosity can enhance soft fascination in connection with the environmental features and that skills in detachment can promote a sense of being away, above and beyond what the environment alone supports. Thus, with gradual training in a mindfulness course, meditation in nature should eventually help restore adaptive resources.

We developed a 5-week course based on open monitoring mindfulness practices and restorative environments theory, called restoration skills training (ReST). The course involved weekly classes in a garden environment, with exercise instructions aimed to stimulate participants' effortless, restorative transactions with the environment through sensory exploration and practice in curiosity and detachment. The course also entailed daily homework. ReST was compared with a conventional mindfulness training (CMT) course held indoors. The CMT course emphasized focused attention practices because these are dominant in mindfulness courses commonly offered to beginners and are by many thought to drive the attentional enhancements in early stages of meditation training (see e.g., Chiesa et al., 2011; Malinowski, 2013; Tang et al., 2015). CMT otherwise had a structure and requirements for class attendance and homework exercises similar to those of ReST.

The CMT course was not intended to represent MBSR or any other specific pre-existing mindfulness training course. Instead, it was

meant to represent the proposed dominant pathway to attentional enhancements in early stages of meditation training: attention network training through focused attention meditation. The design of the two meditation courses thus allowed us to contrast ReST with the dominant approach to achieve attentional enhancements in early stages of meditation training, in order to test their relative effectiveness and respective assumptions.

We conducted four successive rounds of data collection to develop ReST, organized into two studies. Study 1 covered the development of ReST across the first three rounds of data collection, and Study 2 tested the fully developed ReST course in a single round of data collection. Using both behavioral and self-report measures, each study assessed longer-term outcomes of participation as well as short-term outcomes that reflected on the processes engaged with ReST versus CMT. Lymeus et al. (in preparation) present analyses of the longer-term outcomes on psychological health and functioning of participation in the two courses. This paper specifically considers the attentional outcomes of participation in the ReST and CMT courses, as seen in change across the weeks of the course, as well as in the short-term outcomes of participation in the ReST and CMT classes (Study 1) and in meditation sessions within classes (Study 2).

We hypothesized generalized improvement in attentional performance with both courses (H1). We also expected that ReST participants would show signs of restoration with the classes (Study 1) and meditation sessions (Study 2) as indicated by improvement in attentional performance and emotional stress ratings, whereas CMT participants would show signs of effort as indicated by deterioration in attentional performance and hampered stress reduction. However, in line with Lymeus et al. (2017), we expected that any effects indicative of restoration and effort would appear gradually with increasing mindfulness skills acquired across the five weeks of the course. We thus hypothesized improvement with ReST practice and deterioration with CMT practice toward the end of the course (H2).

2. Study 1

2.1. Methods

2.1.1. Design

In the process of developing the ReST course, new participants were recruited and randomly assigned to ReST and CMT in three successive rounds of data collection with the same design. In each round of data collection, the five weekly ReST and CMT classes were held in the late afternoon, a design element that supports the expectation that the participants would come to each class with some potential for restoration of resources that had been depleted in meeting the days' demands (cf. Hartig & Staats, 2006). The design involved assessments before and after the ReST and CMT classes on weeks 1, 3, and 5 of the courses. Study 1 thus has a mixed design with two between-subjects factors: the three rounds of data collection (Round) and the type of course, ReST or CMT (Course). The design also has two within-subjects factors that represent time on different levels: the week in which assessments were made in connection with classes 1, 3, and 5 (Week), and the time in relation to the class when assessments were collected, before or after the class (Time).

2.1.2. Participants

The different rounds of data collection used the same enrollment procedures. We posted flyers in several areas of a university campus, asking for volunteers for a study about mindfulness training. We particularly stated that we sought students with self-perceived stress or concentration problems but no other major health issues and with little or no meditation experience. Screening included the MINI International Neuropsychiatric Interview (Lecrubier et al., 1997) and excluded those who indicated a history of neuropsychiatric disorder, psychoses, hypomanic or manic episodes or recurring depression, moderate to severe post-traumatic stress symptoms, serious self-harm, or suicide attempts. The screening also excluded those with any current moderate to severe mood or anxiety disorders, suicidal ideation, or ongoing psychological or psychiatric treatment.

Eligible individuals who provided informed consent to participate were stratified by gender and randomly assigned to either ReST or CMT within each round of data collection. Altogether, 101 participants were enrolled (median age = 23 [range 18–61]). Twelve of the assigned participants withdrew from the study before the first class or never showed up; 42 participants (26 women) began the ReST course (round 1 n = 9, round 2 n = 14, round 3 n = 19), and 47 participants (31 women) began the CMT course (round 1 n = 10, round 2 n = 15, round 3 n = 22).

The participants in round 1 of data collection were accommodated in one ReST and one CMT course group. Rounds 2 and 3 enrolled larger numbers of participants. To make the course administration manageable in the latter rounds, after assignment to ReST or CMT the participants self-selected into two groups that met on different weekdays. They could not switch groups during the course. Participation in the courses was free of charge and participants were promised three cinema tickets if they also completed all measurements. The study was approved by the regional ethical review board (registration number: 2013/033).

2.1.3. ReST and CMT classes

In each round of data collection, participants in the ReST classes completed adapted mindfulness exercises in garden settings. Participants in the CMT classes completed conventional mindfulness exercises with an emphasis on focused attention practice indoors with no outside views. Each 90-min ReST and CMT class started and ended with 15–30 min of exercise, so exercise constituted much of the class time. However, the classes additionally included themed discussions, homework consultation, and sometimes an intermediate exercise. The classes built on concepts and skills covered in preceding classes and introduced new exercises. With the development for rounds 2 and 3 of data collection of the ReST course, the exact contents of the respective ReST classes were updated

based on experience from the previous round. The format was thus fairly consistent across the courses and rounds of data collection, while the specific exercises and themes were progressively organized across the courses and the ReST approach to practice was developed across the rounds of data collection. In addition to attending the classes, participants were also asked to complete different circa 15-min long homework exercises on most days between the classes. The maximum length of exercises and total exercise load were thus less than in MBSR and more comparable to those used in some earlier adaptations for people with impairing self-regulation problems (cf. Linehan, 1993; Zylowska et al., 2008).

The model for the ReST exercises took form over the rounds of data collection. It built on the conceptual links between curiosity and fascination, and decentering and being away. Beginning in round 1 with exercises fairly similar to the conventional mindfulness exercises, but with more focus on exploration of experiences of the environment, the meditation instructions were successively developed. The participants were led to explore experiences with a given sensory modality in the different classes (i.e. touch, sound, sight, or smell and taste [calorie and caffeine free tea]). The exercises were also formed into three distinct stages: 1) grounding practice (with open or closed eyes, as preferred) in which participants were directed to gently move and stretch different parts of the body in turn, explore and embrace sensations in their bodies and release any unnecessary tension, welcome any thoughts and emotions to be included in the experience but also allow them to pass in their own time, and finally center on sensations associated with the breath; 2) sensory exploration of given objects (sometimes with closed eyes to prevent dominance of vision over other senses), intended to stimulate curiosity and provide experience with different ways to connect with objects; and 3) open sensory exploration of the environment (with open eyes).

2.1.4. Course and assessment settings

The assessments before both ReST and CMT classes, the CMT classes, and the assessments after CMT classes were held in a university campus building. The ReST classes and assessments after ReST classes were held in an adjacent botanical garden located in a moderately dense urban area.

2.1.4.1. Classroom settings. We used several similar, small classrooms with desks and chairs along one side of the room. Assessments were completed at the desks, and the CMT classes were held in a circle near the center of the same room. Some participants sat on the floor but most chose to sit on chairs.

The rooms had closed curtains to prevent outside views and no artwork or decorations. They also had acoustic tile ceilings, soft fluorescent lighting, and off-white wall color. The auditory environment was generally quiet, given the late hour of the workday; however, outside sounds occasionally emanated from traffic, passers-by in the hallway, and other activities in the building. Building on pilot studies (Kihlberg, 2012; Nordfeldt, 2012), the rooms were expected to be low in both restorative qualities and demands for directed attention, and therefore thought to neither support nor interfere with mindfulness practice.

2.1.4.2. Garden settings. Most ReST classes used a tropical greenhouse which in pilot studies had shown high perceived restorative quality compared to different indoor campus settings (i.e., higher ratings of fascination and being away; Kihlberg, 2012; Nordfeldt, 2012). Outdoor areas of the botanical garden and an orangery were also used. These settings, too, were expected to be high in restorative quality and so capable of supporting mindfulness. For unguided exercise, ReST participants were free to move around in all rooms in the greenhouse and to go to the garden outside. They could walk around or find a place to sit as they preferred.

Three larger rooms of the tropical greenhouse each had the necessary amenities and were used for most sitting and standing guided exercise and theoretical discussions: a mildly temperate, stone-paved hall with café style chairs and tables dispersed among canopy trees and flowering plants (16–23 °C, 60–80% relative humidity [RH]); an oblong room with winding gravel paths through dense rainforest vegetation with overhanging foliage and a small terrace with chairs (20–25 °C, 70–90% RH); and a spacious, rounded room dominated by a circular water lily pond that had a small terrace for seating at one end and tropical trees and plants lining the walls (23–27 °C, 70–90% RH). The greenhouse also has three smaller rooms that participants could enter during unguided exercises: an orchid room, a succulent room, and a reception area. Tropical frogs roam freely in the greenhouse, and the several water bodies accommodate small fish. The sounds of water drops hitting water surfaces, leaves, and stone, and the occasional chirping calls of the frogs contribute to the auditory environment of the greenhouse. The climate system occasionally introduces mechanical and background sounds of relays, vents, and fans. Some outside sounds emanating from the weather, traffic, and passers-by could also be heard.

At a different site in the botanical garden, an imposing neoclassical building houses an oblong, stone-paved orangery for cold-season storage of exotic trees in large pots and wood crates. The room has a high ceiling and full-height windows overlooking a water pond and turf with dispersed trees outside. The building also has a grand reception hall. Outside areas used during classes included a coniferous tree grove, a historic kitchen garden, and the different spots that participants sought out themselves for unguided exercises.

The stone-paved hall and reception area of the greenhouse and the grand reception hall of the orangery building – whichever was convenient given the class setting on each occasion – were used for assessments after ReST classes.

2.1.5. Measures

The participants in Study 1 provided attentional performance measures with the Letter–Digit Substitution Test (LDST; van der Elst, van Boxtel, van Breukelen, & Jolles, 2006). The substitution test was chosen over technologically sophisticated assays because of its robust psychometric properties, high change sensitivity, and suitability for brief group administration. Substitution tests are sensitive to variations in the selective and executive components of attention (Davis & Pierson, 2012; Lezak, 2004; Mirsky, Anthony,

Duncan, Ahearn, & Kellam, 1991). LDST has previously shown reliably different effects of mindfulness practice or rest sessions with nature stimuli versus mindfulness practice sessions without nature stimuli (Lymeus et al., 2017). The participants were instructed to quickly and accurately fill in digits (1–9) in blank spaces corresponding to randomly assigned letters, guided by a code-key. The score before the classes is the number of correct responses completed in 60 s, on the last of three consecutive trials. Multiple trials were used before the classes to control familiarity and training effects. The score after the classes was obtained with a single trial to minimize disruption and time demand. All trials used new versions, counterbalanced across participants.

Participants rated their present emotional state with six items from the Swedish Core Affect Scale (SCAS; Västfjäll, Friman, Gärling, & Kleiner, 2002). SCAS items are pairs of negative and positive adjectives that mark opposite poles on a nine-point scale. Participants indicated how they felt with regard to each adjective pair, with 1 meaning complete agreement with the negative adjective, 5 meaning neutral, and 9 meaning complete agreement with the positive adjective. Three adjective pairs belong to each of two affective dimensions: unpleasant activation – pleasant deactivation (e.g., tense – relaxed) on which a score increase corresponds to experienced stress reduction, and unpleasant deactivation – pleasant activation (e.g., indifferent – engaged) on which a score increase corresponds to experienced increase in positively valenced energy. Observed internal consistency (Cronbach's alpha) of the three items in each of the two subscales across all 36 SCAS administrations was, on average for unpleasant activation – pleasant deactivation 0.82 (SD = 0.118; range 0.43–0.98) and for unpleasant deactivation – pleasant activation 0.76 (SD = 0.159, range 0.12–0.92; the one unexpectedly low value [0.12] was obtained for CMT participants on the post-class assessment on week 5, apparently due to very low variation in the responses to one of the constituent items).

2.1.6. Procedures

Participants arriving for ReST and CMT classes were guided as a group to a classroom where they sat down at desks. A researcher gave verbal instructions for each measure. The participants first completed the three trials with LDST. They then completed SCAS. They then proceeded to attend their ReST or CMT class. ReST participants walked up to about 100 m to the class environment.

The first author, with several years' experience in clinical psychology and meditation instruction, led the classes together with different psychologists-in-training in the different rounds of data collection. A preliminary script for the exercises and themed discussions was prepared for each class to ensure consistency.

After their classes, participants again sat down at desks, the ReST participants now in or near the given garden environment. A researcher gave verbal instructions for each measure. The participants completed one trial with LDST followed by SCAS.

2.1.7. Statistical analyses

Our between-subjects design encompassed ReST and CMT courses in three rounds of data collection and course development. Participants were not randomly assigned to the three rounds of data collection; any effects of round could potentially be confounded with external variables such as seasonality or history. We pooled data from the three rounds and included Round as a between-subjects factor in the repeated-measures analyses to disentangle potential effects of Round from effects of the ReST and CMT training and to achieve adequate power. However, because our primary concern here is for effects of Course, Week, and/or Time, we only report main and interaction effects of Round when they would change any conclusions drawn regarding effects of Course, Week, and/or Time uncovered in a given statistical test. Our within-subjects design entailed assessments before and after classes on weeks 1, 3, and 5 of the ReST and CMT courses. Our main hypotheses concerned change across the weeks of training in the courses, from before to after classes, and emerging differences in change from before to after classes across the weeks of the course.

We analyzed our attention assay (LDST) in four steps. In the first step we examined differences between the course groups at the very first measurement point. In the second step we examined change in general attentional capability across the weeks of the course, looking only at LDST data obtained before the classes on weeks 1, 3 and 5. For this we used repeated-measures ANOVA (RM-ANOVA) which included Course (two levels, ReST and CMT) and Round of data collection (three levels) as between-subjects factors and Week (three levels) as a within-subjects factor. The third step in analysis related to effort and restoration within the classes across the weeks of the courses. For this, we extended the RM-ANOVA with a second within-subjects factor; it thus included both Week (three levels) and Time (two levels, before and after classes) as within-subjects factors. The fourth step sought to establish whether any differences in class effects between the groups were independent of differences in levels before the classes. For this, we contrasted the scores obtained after the classes in ANCOVA that included scores obtained before the classes as covariates (see e.g., Vickers & Altman, 2001).

For the emotional state (SCAS) data, we used the same approach to analysis as with the LDST data.

Data were included from all participants who completed the assessments at the different time points included in the respective analyses. Across the repeated assessments before and after classes on weeks 1, 3, and 5, 51 participants provided complete data and could be included in the RM-ANOVA. Given that our focus here is on the effects of participation in the courses and classes rather than on estimating potential treatment outcomes in the population, we consider intention to treat analyses inappropriate for our purposes.

2.2. Results

2.2.1. Initial attention performance

On the initial test with LDST before the first class, the ReST and CMT participants performed similarly (F[80] = 1.36, p = .248, η_p^2 = 0.017). The means fall slightly below the level that could be expected from the modal demographic characteristics of the sample (ReST M = 37.51 [SD = 5.41]; CMT M = 39.91 [SD = 7.28]; cf. van der Elst et al., 2006).

Table 1
Observed means (standard deviations) for performance on the Letter-Digit Substitution Test (LDST) and ratings with the Swedish Core Affect Scale of unpleasant activation – pleasant deactivation (SCAS UA-PD) and unpleasant deactivation – pleasant activation (SCAS UD-PA) in Study 1. The assessments were obtained before the restoration skills training (ReST) and conventional mindfulness training (CMT) classes on weeks 1, 3, and 5 of the course. Repeated-measures ANOVA test statistics are also shown, for the trends across the weeks and their interaction with Course. Descriptive and test statistics are for those participants who provided complete data across all time points (ReST n = 25, CMT n = 26).

Study 1		Week			Repeated-measures ANOVA			
		Week 1	Week 3	Week 5	Contrast	F(1,45)	*p*	η_p^2
LDST	ReST	39.08 (5.66)	41.28 (7.05)	39.92 (6.14)	Week (linear)	11.27	.002	0.200
					Week (quadratic)	7.32	.010	0.138
	CMT	40.58 (7.69)	42.12 (6.83)	43.27 (6.50)	Course × Week (linear)	2.65	.111	0.056
					Course × Week (quadratic)	1.34	.254	0.029
SCAS UA-PD	ReST	4.89 (1.45)	5.12 (1.76)	5.37 (2.00)	Week (linear)	6.24	.016	0.122
					Week (quadratic)	0.63	.433	0.014
	CMT	4.53 (1.59)	5.24 (1.17)	5.53 (1.38)	Course × Week (linear)	0.11	.744	0.002
					Course × Week (quadratic)	0.47	.499	0.010
SCAS UD-PA	ReST	6.53 (1.29)	5.49 (1.69)	5.56 (1.69)	Week (linear)	2.95	.093	0.061
					Week (quadratic)	1.90	.175	0.041
	CMT	6.08 (1.38)	5.90 (1.16)	5.83 (1.20)	Course × Week (linear)	2.99	.091	0.062
					Course × Week (quadratic)	2.12	.152	0.045

Note: Higher scores indicate better LDST performance (observed range 24–59) and more positive emotional state ratings with the SCAS (possible range 1–9, observed range for UA-PD 1.00–8.33 and for UD-PA 1.00–9.00).

2.2.2. Attention performance change with the meditation courses

Table 1 shows descriptive statistics and RM-ANOVA test statistics for the assessments obtained before the classes on weeks 1, 3, and 5 of the courses. As anticipated in H1, the LDST scores obtained before the classes increased across the weeks on average for the three rounds of the study. The average improvement from week 1 to week 5 was rather small: 1.78 (SD = 4.09) letter-digit matches within 60 s, or d = 0.27. The entire total increase was seen between weeks 1 and 3, with 1.86 (SD = 3.91) points improvement, after which performance leveled off between weeks 3 and 5. The change across the weeks was not reliably different among ReST and CMT participants. The average trend was thus for improved LDST performance with both courses, with the bulk of improvement between week 1 and 3. However, the results should be interpreted cautiously because the patterns of change were inconsistent across the rounds of data collection as indicated by significant interactions of Round × Course × Week (for the linear trend, F[1,45] = 4.78, p = .013, η_p^2 = 0.175; and the quadratic trend, F[1,45] = 4.22, p = .021, η_p^2 = 0.158).

2.2.3. Attention performance change with the meditation classes

We extended the RM-ANOVA with the second within-subjects factor (Time, before and after classes). On average across the measures obtained on weeks 1, 3, and 5, participation in the meditation classes was attended by increase in LDST scores (Time F[1,45] = 28.23, p < .001, η_p^2 = 0.385), which was similar in ReST and CMT (Course × Time F[1,45] = 1.22, p = .274, η_p^2 = 0.026). However, the change with the classes differed between ReST and CMT participants across classes 1, 3, and 5, for the quadratic trend (Course × Week × Time F[1,45] = 5.15, p = .028, η_p^2 = 0.103); the change patterns apparently diverged on week 5 when ReST participants but not CMT participants appeared to improve further. ANCOVAs controlling for the LDST score before the class showed that change with the given class was similar for ReST and CMT participants on week 1 and week 3. However, ReST participants' LDST scores improved more with class 5 than CMT participants'. Table 2 shows descriptive statistics and ANCOVA test statistics for group differences in change with the respective classes. Thus, in line with H2, ReST practice was attended by greater short-term attentional benefit than CMT practice toward the end of the courses, although CMT participants' performance never deteriorated substantially from the initial value.

2.2.4. Emotional change with the meditation classes

Before the meditation classes, participants on average rated their emotional state as being close to center on the scale with regard to the dimensions of unpleasant activation – pleasant deactivation and unpleasant deactivation – pleasant activation (see Table 1). The ratings also increased across the weeks of the course. The participants were thus not particularly unpleasantly activated (e.g., tense) or unpleasantly deactivated (e.g., indifferent) and so had limited room for improvement with the meditation practice, increasingly less so across the weeks. Yet, participation in the classes was attended by an average shift of about 1.5 scale points toward greater pleasant deactivation (e.g., relaxation) (Time F[1,45] = 82.73, p < .001, η_p^2 = 0.648), which was similar for ReST and CMT participants (Course × Time F[1,45] = 0.13, p = .722, η_p^2 = 0.003) and across the weeks (Course × Week × Time linear trend F[1,45] = 0.72, p = .399, η_p^2 = 0.016; and quadratic trend F[1,45] = 0.35, p = .556, η_p^2 = 0.008). In contrast, participation in the classes was not attended by any significant change in ratings of unpleasant deactivation – pleasant activation (Time F[1,45] = 1.14, p = .291, η_p^2 = 0.025; Course × Time F[1,45] = 0.36, p = .551, η_p^2 = 0.008; Course × Week × Time linear trend F[1,45] < 0.01, p = .947, η_p^2 < 0.001; Course × Week × Time quadratic trend F[1,45] < 0.01, p = .955, η_p^2 < 0.001). The emotion ratings thus show that both ReST and CMT participants were pleasantly deactivated with the meditation classes.

Table 2
Estimated means (standard errors) for change with the restoration skills training (ReST) and conventional mindfulness training (CMT) classes in Letter-Digit Substitution Test (LDST) scores and ratings with the Swedish Core Affect Scale of unpleasant activation – pleasant deactivation (SCAS UA-PD) and unpleasant deactivation – pleasant activation (SCAS UD-PA) in Study 1. The assessments were obtained before and after the ReST and CMT classes on week 1, 3, and 5 of the course. Between-subjects ANCOVA test statistics are also shown, for tests that included the pre-session scores as a covariate. Descriptive and test statistics comprise those participants who provided data on the relevant assessments on the respective weeks (week 1: ReST n = 40, CMT n = 45; week 3: ReST n = 33, CMT n = 31; week 5: ReST n = 29, CMT n = 31).

Study 1		Course		Between-subjects ANCOVA			
		ReST	CMT	Contrast	F	p	η_p^2
LDST change	Week 1	2.00 (0.56)	0.98 (0.53)	Course	1.69	.197	0.020
				Pre-class score	1.98	.163	0.024
	Week 3	1.32 (0.73)	2.60 (0.76)	Course	1.47	.230	0.024
				Pre-class score	3.64	.061	0.056
	Week 5	2.74 (0.85)	0.25 (0.82)	Course	4.22	.045	0.069
				Pre-class score	0.20	.655	0.004
SCAS UA-PD change	Week 1	1.88 (0.25)	1.41 (0.23)	Course	1.92	.170	0.023
				Pre-class score	61.00	< .001	0.427
	Week 3	1.28 (0.25)	1.23 (0.25)	Course	0.01	.905	< 0.001
				Pre-class score	16.55	< .001	0.225
	Week 5	0.77 (0.28)	1.32 (0.26)	Course	2.05	.158	0.037
				Pre-class score	12.96	.001	0.196
SCAS UD-PA change	Week 1	−0.02 (0.20)	−0.48 (0.20)	Course	2.62	.110	0.032
				Pre-class score	26.26	< .001	0.252
	Week 3	0.25 (0.26)	0.52 (0.25)	Course	0.59	.447	0.010
				Pre-class score	28.32	< .001	0.332
	Week 5	0.32 (0.25)	0.44 (0.24)	Course	0.12	.728	0.002
				Pre-class score	29.27	< .001	0.356

Note: Higher change scores indicate improvement in attention performance for the LDST (observed range: −9 to 11) and larger shift to a more positive emotional state for the SCAS (possible range: −8 to 8; observed range: for UA-PD, −7.33 to 6.00, and for UD-PA, −4.33 to 6.33). The error degrees of freedom for the different tests vary with the number of included participants: for week 1 = 82, week 3 = 61, and week 5 = 57).

2.3. Discussion

The results of Study 1 indicate that both ReST and CMT were associated with generalized improvement in attention performance. Further, attendance in the ReST classes was consistently restorative with regard to attention performance and self-reported pleasant deactivation across the weeks of the course and the rounds of data collection. In contrast, participants in the CMT classes showed decreasing benefit from the classes with regard to attention performance toward the end of the course. A difference thus emerged with the fifth class, in line with expectations that differences in practice effects might be driven by increasing skill acquired across the weeks of training in the different approaches to mindfulness. However, contrary to expectations, the benefits of ReST did not clearly increase and CMT participants' attention performance never deteriorated substantially from the levels seen before the classes.

The results of Study 1 are thus encouraging but only partially in line with our hypotheses. The failure to confirm increasing benefit with ReST and attentional effort and hampered emotional stress reduction with the later CMT classes could be due to one or more methodological problems. First, data collection was done over three rounds of the courses across which ReST was developed and across which participants could not be randomly assigned. Statistical control for round may not fully remedy the potential confounds introduced. Second, the class environments and settings used for assessments were not consistent for the classes across the weeks of ReST and in the different rounds of data collection. Also, CMT participants unlike ReST participants had classes in the same room where they had completed assessments before the classes. Third, assessments used to study longer-term change with the courses were obtained before the classes in the course but lacked a pre-course measurement point. Fourth, measures were obtained before and after 90-min classes that entailed both meditation exercise and other content. The classes also had different specific meditation exercises and themed discussions on the different weeks and in the different rounds of data collection. Change with the classes was thus confounded with the specific choices of exercises, course progression across the weeks, and course development. Fifth, attention was assessed only with the LDST, which is sensitive to variation in both selective and executive attention and thus rather unspecific.

3. Study 2

Study 2 aimed to replicate Study 1 with several methodological improvements. We tested the fully developed ReST course against CMT in a single round of data collection. We used the same meditation settings across the weeks with the respective courses, and we had CMT participants, like ReST participants, change rooms when the assessments before meditation were completed. We also included a pre-course measurement point that could be used to familiarize and train participants with the attention tests as well as allow disentanglement of residual training effects from the specific change once the course started. Given our primary concern for effort and restoration incurred with meditation, rather than with the effects of participation in entire classes which entail different exercises as well as other content, measures were obtained before and after isolated meditation sessions with consistent exercise

instructions across the weeks. Given our concern for development in practice effects with increasing skills over weeks of mindfulness training, we also gave the same exercise used with the assessments as daily homework every week. To further strengthen confidence and specificity in the conclusions about attention effects of the courses and meditation sessions, we included together with the LDST an additional attention assay that specifically taps executive capabilities.

3.1. Methods

3.1.1. Design

The design involved repeated assessments obtained at a pretest and then before and after meditation sessions on weeks 1, 3, and 5 of the ReST and CMT courses. We thus have a mixed design with one between-subjects factor (Course; ReST versus CMT) and two within-subjects factors that represent time on different levels: assessments at pretest and then on weeks 1, 3, and 5 of the course (Week), and before and after meditation sessions on weeks 1, 3, and 5 (Time). The procedures were again completed in the late afternoon so that participants would more likely have a potential for restoration.

3.1.2. Participants

Recruitment procedures were identical to those in Study 1. Eligible participants who provided informed consent to participate (N = 59; median age = 23 [range 18–49]) were stratified by gender and randomly assigned to either ReST or CMT. After assignment but before the first class, eight participants withdrew from the study; 27 participants (19 women) began the ReST course, and 24 participants (18 women) began the CMT course.

Once assigned to ReST or CMT, participants could choose between three course groups that met on different weekdays. They could not switch groups during the course. Participation in the courses was free of charge and participants were promised three cinema tickets if they also completed all measurements.

3.1.3. ReST and CMT meditation sessions

ReST participants completed an open monitoring mindfulness exercise, adapted to enhance contact with potentially restorative features of the environment. In contrast, CMT participants completed a conventional focused attention mindfulness exercise indoors with no outside views. The 20-min meditation sessions were held at the start of the ReST and CMT classes on weeks 1, 3, and 5 of the courses. The course leaders gave the same scripted practice instructions every time. The participants also received a recording of the same so-called “basic exercise” and were instructed to complete it as formal homework on most days of each week of the course. For ReST, the meditation session comprised (1) grounding in bodily sensations and the breath with open or closed eyes as preferred, (2) opening up to sensory impressions from the environment (tactile, auditive, and visual, in turn) and private experiences, and (3) open monitoring of the experiential stream from moment to moment. For CMT, it comprised (1) grounding in bodily sensations; and (2) mindfulness of breath practice including noticing, accepting, and letting go of any external or private experiences that caught attention. CMT participants received instructions to keep their eyes gently closed during the entire exercise. Both ReST and CMT participants remained seated throughout the exercise. (Contact the first author to discuss possibilities for using our scripts in replications.)

3.1.4. Environments

Different small classrooms were used for the assessments before the meditation sessions (see Section 2.1.4.1). The ReST participants completed their meditation sessions and the assessments after the sessions sitting dispersed among canopy trees and flowering plants on café style chairs in the stone-paved hall of the greenhouse (see Section 2.1.4.2). The CMT participants completed their meditation sessions and the assessments after the sessions in a classroom at the other end of the building from the room used for assessments before the sessions, so that their movement from the pre-session assessment setting to the course setting was comparable to that made by the participants in the ReST course.

3.1.5. Measures

In addition to the LDST and SCAS used in Study 1, the participants also completed a version of the Trail-Making Test (TMT; Lezak, 2004; Reitan, 1958; Tombaugh, 2004). In part A of the TMT, participants were to draw a line between consecutive numbers in circles spread out pseudo-randomly on a sheet of paper (i.e. place and order in a grid determined by random number generator and then minimally rearranged to ensure straight lines could be drawn in correct sequence). In part B, they saw numbered and lettered circles and drew a line switching between the numerical and alphabetical sequence (i.e. 1-A-2-B, etc.). They received instructions to work as quickly as they could without making mistakes. TMT-A assesses information processing speed while TMT-B adds a set-switching component that relies on executive capabilities. The intra-individual difference between TMT-A and TMT-B performance on a given trial shows the performance decrement incurred by the set-switching task. That switching cost index thus isolates the executive component from the information processing component. To accommodate brief group administration and repeated measurements, we created several TMT-A and TMT-B versions, matched for length. All participants worked for 20 s on TMT-A and then 20 s on TMT-B. Three scores were obtained: the number of correct connections completed on TMT-A and TMT-B (possible range 0–25), respectively, and the switching cost conceived as TMT-A minus TMT-B. Higher scores on TMT-A and TMT-B indicate better information processing and executive performance. Lower switching cost indicates better executive functioning relative to the given level of information processing capability within a trial. All trials used new versions, counterbalanced across participants.

Observed internal consistency (Cronbach’s alpha) across the 12 SCAS administrations was, on average for unpleasant activation –

pleasant deactivation 0.88 (SD = 0.05; range 0.79–0.95) and for unpleasant deactivation – pleasant activation 0.79 (SD = 0.09, range 0.63–0.89).

3.1.6. Procedures

At the meeting for enrollment in the study, participants completed three trials with the LDST followed by three trials with the TMT (with part A followed by part B within each trial). The first of these were practice runs. When the course started, participants arriving for ReST or CMT classes were guided as a group to a classroom where they sat down at desks. A researcher gave verbal instructions for each measure. The participants completed two trials with the LDST, two trials with the TMT, and then the SCAS. They then moved location to attend their ReST or CMT class. The ReST participants walked about 50 m to the tropical greenhouse and CMT participants walked a comparable distance to the room at the other end of the campus building.

Different instructors led the meditation sessions and the subsequent ReST and CMT classes on three different weekdays. The same scripted instructions were used for the meditation sessions every time. A preliminary script was also prepared for the other exercises and themed discussions of each class to ensure quality and consistency across instructors. The first author led the meditation sessions and classes on one of the weekdays and two psychologists-in-training with previous meditation experience sat in on those classes to observe and assist. The psychologists-in-training then led the meditation sessions and classes on the two remaining weekdays.

As participants sat down for their class on weeks 1, 3, and 5 of the course, the instructor welcomed them and immediately commenced with instructions for the meditation sessions. After 20 min of meditation, the participants moved their chairs to tables in the same room, where they completed the LDST, TMT, and SCAS. They moved the chairs back to participate in the rest of the class, which lasted 90 min in total.

3.1.7. Statistical analyses

The steps of the statistical analyses were similar to those in Study 1 with some exceptions. The RM-ANOVA only had one between-subjects factor (Course; ReST vs. CMT). The RM-ANOVA used to examine change in LDST and TMT scores across the weeks of the study had four levels of the within-subjects factor (enrollment and weeks 1, 3, and 5). For the extended RM-ANOVA that analyzed change with the meditation sessions, the enrollment scores were omitted from the within-subjects model which thus only had three levels of the factor Week (weeks 1, 3, and 5) together with the two-level factor Time (before and after meditation sessions).

Because the LDST and TMT scores obtained in the two consecutive trials (following the practice run) used at enrollment and before the meditation sessions did not show any significant within-subjects change in Study 2, we used the average of those two trials as the score before sessions in order to reduce error variance. For the ANCOVA for change with the individual meditation sessions in TMT switching cost, any significant course differences were checked with an additional ANCOVA with switching cost after the session entered as the dependent variable and the TMT-A score after the session as a covariate, to check whether any difference in switching cost (i.e. TMT-A minus TMT-B) was only due to a difference in TMT-A performance. All analyses which indicated a relevant significant effect were also repeated with the different instructors entered as an additional between-subjects factor (tree levels) together with Course. However, we do not report the results of those analyses here because Instructor never showed a significant main or interactive effect with any outcome and because we consider the effects of the Instructor factor unreliable due to the small n in the individual course groups. Across the repeated assessments at enrollment and before and after meditation sessions on weeks 1, 3, and 5, 29 participants provided complete data and could be included in the RM-ANOVA.

3.2. Results

3.2.1. Initial attention performance

As in Study 1, in the initial test with LDST at the enrollment, the participants assigned to ReST and CMT, respectively, performed slightly below the level expected given the modal demographic characteristics of the sample (ReST M = 37.41 [SD = 6.03]; CMT M = 37.82 [SD = 4.71]; F[56] = 0.08, p = .775, η_p^2 = 0.001). They thus showed limited objective signs of attention problems as assessed with the LDST.

The version we used of TMT has no normative reference data and so must be considered only in terms of the observed scores and variability. For TMT-A, the participants completed about 14 correct connections on average in the 20 s (ReST M = 13.93 [SD = 3.25]; CMT M = 13.82 [SD = 2.64]; F[56] = 0.02, p = .886, η_p^2 < 0.001. For TMT-B, they completed about 10 correct connections (ReST M = 9.71 [SD = 2.40]; CMT M = 9.67 [SD = 1.86]; F[56] = 0.07, p = .933, η_p^2 < 0.001). The average switching cost (TMT-A – TMT-B score) was thus, for ReST M = 4.21 (SD = 2.53) and for CMT M = 4.15 (SD = 2.35) (F[56] = 0.01, p = .920, η_p^2 < 0.001) or a performance decrement of about 30%.

3.2.2. Attention performance change with the courses

Table 3 shows descriptive statistics and RM-ANOVA test statistics for the assessments obtained at the enrollment and before the meditation sessions on weeks 1, 3, and 5 of the courses. The LDST scores increased across the weeks. The average improvement from enrollment to week 5 was medium sized: 4.83 (SD = 3.81) letter-digit matches within 60 s, or d = 0.60. As in Study 1 and in support of H1, the bulk of the improvement (2.78 [SD = 3.87] letter-digit matches or 58% of the total mean increase) again occurred between weeks 1 and 3. The change across the weeks in these assessments was not reliably different among ReST and CMT participants.

As with the LDST scores, both the TMT-A and TMT-B scores obtained before the sessions increased on average from the pretest at enrollment across weeks 1, 3, and 5. The average improvement from enrollment to week 5 was medium sized for TMT-A: 1.40 (SD = 2.97) points, or d = 0.43; and medium to large for TMT-B: 1.33 (SD = 2.35) points, or d = 0.70. For TMT-A, the increase was

Table 3
Observed means (standard deviations) for performance on the Letter-Digit Substitution Test (LDST) and Trail-Making Test (TMT) and ratings with the Swedish Core Affect Scale of unpleasant activation – pleasant deactivation (SCAS UA-PD) and unpleasant deactivation – pleasant activation (SCAS UD-PA) in Study 2. The assessments were obtained at the time of enrollment in the study (LDST and TMT only) and then before the restoration skills training (ReST) and conventional mindfulness training (CMT) classes on weeks 1, 3, and 5 of the course. Repeated-measures ANOVA test statistics are also shown, for the trends across the weeks and their interaction with Course. Descriptive and test statistics are for those participants who provided complete data across all time points (ReST n = 15, CMT n = 14).

Study 2		Week				Repeated-measures ANOVA			
		Pretest	Week 1	Week 3	Week 5	Contrast	F(1,27)	p	η_p^2
LDST	ReST	37.37 (6.98)	38.37 (6.78)	41.40 (7.58)	42.47 (6.16)	Week (linear)	48.88	< .001	0.644
						Week (quadratic)	< 0.01	.977	< 0.001
						Week (cubic)	2.74	.110	0.092
	CMT	36.96 (5.29)	38.04 (6.34)	40.54 (7.99)	41.50 (7.52)	Course × Week (linear)	0.20	.655	0.008
						Course × Week (quadratic)	0.02	.900	0.001
						Course × Week (cubic)	0.06	.808	0.002
TMT-A	ReST	14.27 (2.93)	13.93 (2.24)	14.17 (2.34)	14.80 (4.07)	Week (linear)	6.79	.015	0.201
						Week (quadratic)	0.89	.354	0.032
						Week (cubic)	0.05	.827	0.002
	CMT	13.36 (2.38)	14.00 (2.40)	14.57 (2.78)	15.68 (3.30)	Course × Week (linear)	2.52	.124	0.085
						Course × Week (quadratic)	0.11	.743	0.004
						Course × Week (cubic)	0.15	.702	0.006
TMT-B	ReST	9.93 (1.65)	9.63 (1.80)	10.90 (1.74)	11.33 (2.34)	Week (linear)	14.82	.001	0.354
						Week (quadratic)	1.56	.222	0.055
						Week (cubic)	7.71	.010	0.222
	CMT	9.54 (1.81)	8.96 (1.99)	10.50 (2.97)	10.79 (1.73)	Course × Week (linear)	< 0.01	.949	< 0.001
						Course × Week (quadratic)	0.01	.923	< 0.001
						Course × Week (cubic)	0.21	.648	0.008
TMT switching cost	ReST	4.33 (2.85)	4.30 (2.28)	3.27 (1.37)	3.47 (3.33)	Week (linear)	0.12	.736	0.004
						Week (quadratic)	0.01	.934	< 0.001
						Week (cubic)	4.14	.052	0.133
	CMT	3.82 (2.23)	5.04 (2.59)	4.07 (2.63)	4.89 (2.59)	Course × Week (linear)	2.09	.160	0.072
						Course × Week (quadratic)	0.11	.746	0.004
						Course × Week (cubic)	0.32	.575	0.012
SCAS UA-PD	ReST		4.96 (1.79)	4.82 (1.77)	4.20 (1.42)	Week (linear)	0.31	.580	0.011
						Week (quadratic)	1.16	.291	0.041
	CMT		5.00 (1.88)	5.55 (1.22)	5.26 (1.93)	Course × Week (linear)	1.33	.259	0.047
						Course × Week (quadratic)	0.08	.781	0.003
SCAS UD-PA	ReST		6.42 (1.59)	6.36 (1.68)	5.67 (1.95)	Week (linear)	0.78	.384	0.028
						Week (quadratic)	0.39	.537	0.014
	CMT		6.19 (1.37)	6.36 (1.52)	6.38 (1.31)	Course × Week (linear)	2.19	.150	0.075
						Course × Week (quadratic)	0.15	.698	0.006

Note: Higher scores indicate better attention performance for LDST (observed range 21.00–54.50), TMT-A (possible range 0–25, observed range 6.00–22.50, and TMT-B (possible range 0–25, observed range 3.50–16.00). Lower scores for TMT switching cost (observed range −4.00 to 12.00) indicate better ability to handle the set-switching task in TMT-B relative to simply connecting a single series of targets in TMT-A. Higher scores indicate more positive emotional state ratings with the SCAS (possible range 1–9, observed range for UA-PD 1.67–8.33 and for UD-PA 2.00–9.00).

fairly evenly distributed across the weeks. For TMT-B, however, the entire total increase was seen between week 1 and 3 (1.40 [*SD* = 1.72]) whereafter performance leveled off. The change across the weeks was not reliably different between ReST and CMT participants. Furthermore, the switching cost – our primary assay of short-term change in executive attention – did not change reliably from enrollment across the pre-session measures obtained on weeks 1, 3, and 5. The change in switching cost across the weeks was also not reliably different among ReST and CMT participants.

3.2.3. Attention performance change with the meditation sessions

We excluded the pretest and extended the RM-ANOVA with the second within-subjects factor (Time, before and after meditation sessions). As in Study 1, participation in the ReST and CMT meditation sessions was attended by comparable increase in LDST score on average (Time $F[1,27] = 18.34$, $p < .001$, $\eta_p^2 = 0.404$; Course × Time $F[1,27] = 3.21$, $p = .085$, $\eta_p^2 = 0.106$). However, as shown in Fig. 1, the change with the sessions again showed different patterns across weeks 1, 3, and 5 for the ReST and CMT participants (Course × Week × Time linear trend $F[1,27] = 9.02$, $p = .006$, $\eta_p^2 = 0.250$). The ANCOVA for differences in change in LDST scores from before to after the sessions controlling for scores before the sessions showed that ReST and CMT participants improved similarly on week 1 and week 3. As in Study 1 and in line with H2, however, the change differed on week 5 when ReST participants' LDST scores increased while CMT participants' scores decreased. Table 4 shows descriptive statistics and ANCOVA test statistics for group differences in change with the respective meditation sessions.

With the LDST results in line with our expectations, we sought to confirm a similar pattern in the TMT results. In contrast to LDST performance, participation in the ReST and CMT meditation sessions was not attended by improvement in TMT switching cost on

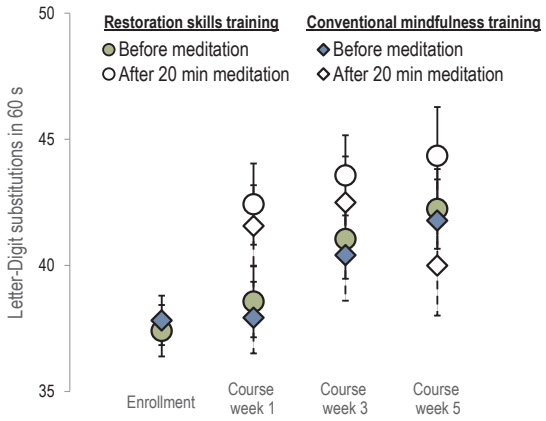


Fig. 1. Letter-Digit Substitution Test performance at enrollment in the study, and then before and after 20-min meditation sessions on weeks 1, 3, and 5 of the restoration skills training (ReST) and conventional mindfulness training (CMT) courses. The same exercise instructions were used for the meditation sessions on each of the weeks and participants trained with the same exercises as formal homework between the classes. *Note:* Higher scores indicate better LDST performance (observed range 18–58). The figure is based on data from the 29 participants who provided complete data for each measurement point displayed.

average (Time $F[1,27] = 1.95$, $p = .174$, $\eta_p^2 = 0.067$; Course × Time $F[1,27] = 0.96$, $p = .337$, $\eta_p^2 = 0.034$). The degree of change differed only marginally between ReST and CMT participants across weeks 1, 3, and 5 (Course × Week × Time linear trend $F[1,27] = 2.90$, $p = .100$, $\eta_p^2 = 0.097$), though with a pattern similar to that seen in the LDST scores as shown in Fig. 2. ANCOVA for differences in change in TMT switching cost with the meditation sessions controlling for switching cost before the sessions showed that it was similar for ReST and CMT participants on week 1 and week 3. As with the LDST scores and in line with H2, however, switching cost improved (i.e. was reduced) with the session on week 5 for ReST but not for CMT participants. We also saw that change with the meditation sessions in TMT-A performance differed between the course groups on week 5, and thus could potentially explain any differences seen in switching cost independent of any real change in executive attention performance. We checked for this by analyzing switching cost after the meditation session with TMT-A performance after the meditation session entered as a covariate. The group difference in switching cost was weakened but held statistically (ReST adjusted $M = 2.63$ [95%CI = 1.39, 3.87]; CMT adjusted $M = 4.62$ [95%CI = 3.29, 5.95]; $F[1,29] = 4.34$, $p = .046$, $\eta_p^2 = 0.130$); the difference in switching cost was thus not only due to differences in post-session TMT-A performance.

Thus, ReST practice was consistently attended by short-term improvement in selective attention, as assessed with LDST, and was attended by improvement in executive attention as assessed with TMT switching cost toward the end of the course. In contrast, CMT practice appeared to be attended by deteriorated selective and executive attention performance toward the end of the course.

3.2.4. Emotional change with the meditation sessions

As in Study 1, participants on average rated their emotional state before the meditation sessions as being close to the center of the scale with regard to the dimensions of unpleasant activation – pleasant deactivation and unpleasant deactivation – pleasant activation (see Table 3). The ratings obtained before the sessions did not change significantly across the weeks. The participants were thus not particularly unpleasantly activated (e.g., tense) or unpleasantly deactivated (e.g., indifferent) and so had limited room for emotional improvement with the meditation practice. Yet, participation in the meditation sessions was attended by an average shift toward greater pleasant deactivation (e.g., relaxation) (Time $F[1,28] = 78.04$, $p < .001$, $\eta_p^2 = 0.736$), more so for ReST participants with 2.08 (95%CI = 1.57, 2.59) scale points improvement than for CMT participants with 1.14 (95%CI = 0.59, 1.68) points improvement (Course × Time $F[1,28] = 6.71$, $p = .015$, $\eta_p^2 = 0.193$). Change in the two groups was consistent across the course weeks (Course × Week × Time linear trend $F[1,28] = 1.10$, $p = .303$, $\eta_p^2 = 0.038$; and quadratic trend $F[1,28] = 0.02$, $p = .890$, $\eta_p^2 = 0.001$). In contrast, participation in the classes was not attended by any significant change in ratings of unpleasant deactivation – pleasant activation (Time $F[1,28] = 0.22$, $p = .644$, $\eta_p^2 = 0.008$; Course × Time $F[1,28] = 1.16$, $p = .290$, $\eta_p^2 = 0.040$; Course × Week × Time linear trend $F[1,28] = 0.58$, $p = .454$, $\eta_p^2 = 0.020$; Course × Week × Time quadratic trend $F[1,28] = 1.56$, $p = .222$, $\eta_p^2 = 0.053$). Partially in line with H2, the emotion ratings thus show that ReST participants reported relatively greater stress reduction with the meditation sessions than the CMT participants, although to a consistent rather than increasing degree.

3.3. Discussion

The participants in Study 2 attended the fully developed ReST course or a CMT course that was closely matched for structure and exercise requirements within classes and for the course as a whole. Looking across the courses, their general attention performance improved comparably in the first weeks, whereafter attentional improvement leveled off. For the meditation sessions, the participants completed the same basic exercise based on the core principles of the different approaches to mindfulness training in ReST and the conventional mindfulness course. They also trained with the same course-specific exercises as homework throughout the course. The results confirmed that the ReST practice of open sensory exploration in a setting rich in natural stimuli was consistently attended by

Table 4
Estimated means (standard errors) for change with the restoration skills training (ReST) and conventional mindfulness training (CMT) classes in Letter-Digit Substitution Test (LDST) and Trail-Making Test (TMT) scores and ratings with the Swedish Core Affect Scale of unpleasant activation – pleasant deactivation (SCAS UA-PD) and unpleasant deactivation – pleasant activation (SCAS UD-PA) in Study 2. The assessments were obtained before and after the ReST and CMT classes on weeks 1, 3, and 5 of the course. Between-subjects ANCOVA test statistics are also shown, for tests that included the pre-session scores as a covariate. Descriptive and test statistics comprise those participants who provided data on the relevant assessments on the respective weeks (week 1: ReST n = 21, CMT n = 21; week 3: ReST n = 21, CMT n = 16; week 5: ReST n = 17, CMT n = 16).

Study 2		Course		Between-subjects ANCOVA			
		ReST	CMT	Contrast	F	*p*	η_p^2
LDST change	Week 1	3.86 (0.79)	3.64 (0.79)	Course	0.04	.846	0.001
				Pre-session score	0.01	.920	0.001
	Week 3	2.54 (0.60)	2.07 (0.69)	Course	0.27	.609	0.008
				Pre-session score	1.03	.318	0.029
	Week 5	2.10 (0.93)	−1.76 (0.96)	Course	8.42	.007	0.219
				Pre-session score	0.53	.471	0.017
TMT-A change	Week 1	−1.02 (0.65)	1.35 (0.65)	Course	6.67	.014	146.
				Pre-session score	2.16	.149	0.053
	Week 3	0.44 (0.75)	1.05 (0.86)	Course	0.28	.600	0.008
				Pre-session score	7.07	.012	0.172
	Week 5	−1.25 (0.61)	1.92 (0.62)	Course	13.29	.001	0.307
				Pre-session score	7.73	.009	0.205
TMT-B change	Week 1	−0.10 (0.56)	1.48 (0.56)	Course	3.85	.057	0.090
				Pre-session score	10.71	.002	0.215
	Week 3	0.59 (0.56)	0.32 (0.64)	Course	0.10	.755	0.003
				Pre-session score	4.13	.050	0.108
	Week 5	0.85 (0.54)	0.58 (0.58)	Course	0.11	.738	0.004
				Pre-session score	3.68	.065	0.113
TMT switching cost change	Week 1	−1.09 (0.80)	0.05 (0.80)	Course	1.00	.325	0.025
				Pre-session score	10.59	.002	0.214
	Week 3	−0.34 (0.78)	0.97 (0.90)	Course	1.19	.283	0.034
				Pre-session score	14.61	.001	0.301
	Week 5	−2.28 (0.69)	1.45 (0.73)	Course	13.37	.001	0.315
				Pre-session score	21.32	< .001	0.424
SCAS UA-PD change	Week 1	2.09 (0.27)	1.71 (0.28)	Course	0.97	.330	0.023
				Pre-session score	13.74	.001	0.251
	Week 3	1.77 (0.31)	1.63 (0.36)	Course	0.09	.766	0.003
				Pre-session score	6.30	.017	0.156
	Week 5	2.08 (0.40)	1.00 (0.42)	Course	3.34	.078	0.100
				Pre-session score	2.62	.116	0.080
SCAS UD-PA change	Week 1	0.32 (0.21)	−0.13 (0.22)	Course	2.07	.158	0.048
				Pre-session score	2.84	.100	0.065
	Week 3	−0.12 (0.26)	0.18 (0.30)	Course	0.54	.469	0.016
				Pre-session score	16.92	< .001	0.332
	Week 5	0.26 (0.26)	−0.01 (0.27)	Course	0.49	.488	0.016
				Pre-session score	13.98	.001	0.318

Note: Higher change scores indicate improvement in attention performance for the LDST (observed range −9.00 to 8.50), TMT-A (observed range −7.00 to 6.50), and TMT-B (observed range −5.00 to 9.50). Negative change scores for TMT switching cost (observed range −10.00 to 7.00) indicate better ability to handle the set-switching task in TMT-B relative to simply connecting a single series of targets in TMT-A. Higher change scores for SCAS indicate larger shift to a more positive emotional state (observed range, for SCAS UA-PD −1.67 to 4.67, and for SCAS UD-PA −2.67 to 3.67). The error degrees of freedom for the different tests vary with the number of included participants: For week 1 = 39, week 3 = 34, and week 5 = 30.

improved attention performance as assessed with the LDST while the CMT mindfulness of breath practice seemed to incur greater effort with increasing training experience. Unlike Study 1, CMT participants' LDST performance after the last of the meditation sessions actually deteriorated from the levels seen before the session. With regard to TMT switching cost, CMT participants consistently showed little change from before to after the sessions while ReST participants improved toward the end of the course, and a benefit for ReST compared to CMT practitioners emerged with the last of the sessions. Consistently across the weeks, the pleasant deactivation ratings improved more with the ReST practice sessions than with the CMT sessions.

The methodological improvements from Study 1 to Study 2 allowed us to more specifically target processes of restoration and effort with the different approaches to mindfulness practice. In addition to the enhanced comparability and consistency of the ReST and CMT courses and exercises mentioned above, we used the same assessment and meditation settings within the course groups across the weeks to eliminate variability caused by changes in settings. The pretest with the attention tests at enrollment allowed us to conclude that improvement in the scores in the first weeks of the course were largely independent of training effects with the tests. We were also able to broaden our operationalization of attention to better represent executive capability.

However, some weaknesses remain. First, we relied on the expectation that the participants on average had a potential for

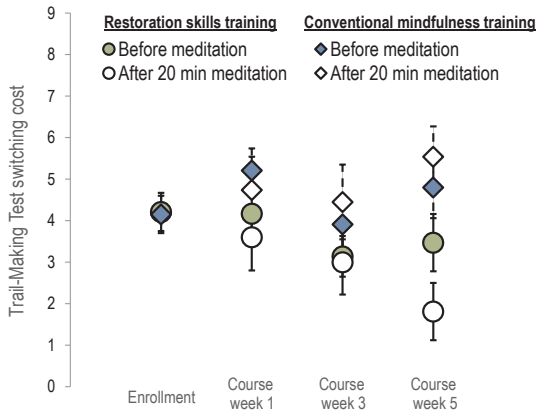


Fig. 2. Trail-Making Test (TMT) switching cost at enrollment in the study, and then before and after 20-min meditation sessions on weeks 1, 3, and 5 of the restoration skills training (ReST) and conventional mindfulness training (CMT) courses. The same exercise instructions were used for the meditation sessions on each of the weeks and participants trained with the same exercises as formal homework between the classes. *Note:* TMT switching cost is the difference between the number of correct connections completed in 20 s on TMT-A versus TMT-B. Lower scores for TMT switching cost (observed range −4 to 12) indicate better ability to handle the set-switching task in TMT-B relative to simply connecting a single series of targets in TMT-A. The figure is based on data from the 29 participants who provided complete data for each measurement point displayed.

restoration with the meditation sessions. Our participants showed only limited objective signs of attention problems at the pretest even though they volunteered on a premise of self-perceived stress or concentration problems; however, the improvements seen with the ReST meditation sessions and with the CMT session on the first week show that they did have at least some room for improvement. Yet, we might have seen more pronounced effects had we incorporated a fatigue induction (Ohly et al., 2016; cf. Hartig & Jahncke, 2017). Second, the study could not be blinded. Any unmeasured allegiance and expectancy effects could thus not be eliminated. However, we ensured that allegiance and expectancy should be relatively high in both course groups by teaching both ReST and CMT participants a practice rationale that entailed a prospect of attentional enhancements, with only the proposed process differing (i.e. restoration in ReST and training in CMT). Also, both ReST and CMT participants volunteered for and received bona fide mindfulness courses (we show elsewhere that the courses had generally comparable longer-term mental health benefits and perceived quality; Lymeus et al., in preparation). Third, attentional effort and restoration with practice could be confounded with conceptually and functionally overlapping processes like motivation. However, the improvements seen with ReST sessions concurred with pleasant deactivation (i.e. relaxation) but not with pleasant activation (i.e. engagement).

4. General discussion

The approach to mindfulness practice with open sensory exploration in a nature-rich environment as used in ReST appears to support beginning meditators with stress or concentration problems better than conventional mindfulness training. Both courses appeared to engender longer-term improvement in attentional capabilities; however, the ReST participants also evidenced short-term restoration of attentional and emotional adaptive resources with the practice. In contrast, the conventional approach to mindfulness training for beginners, and specifically mindfulness of breath practice, seemed to initially restore attentional performance capabilities in the short-term but then increasingly tax attentional resources toward the end of the course. These outcomes are far from trivial because they have specific implications for mindfulness theory and practice.

4.1. Implications for theory and practice

Short-term effort incurred with conventional mindfulness exercises purportedly drives the longer-term attentional improvements seen with regular practice (see, e.g., Chiesa et al., 2011; Malinowski, 2013; Tang et al., 2015). However, the differences between ReST and CMT in restoration and effort seen with the classes and meditation sessions in our two studies seemed to emerge over a number of weeks in training; a pattern similar to that previously reported by Lymeus et al. (2017). Clear signs of effort among CMT participants were seen only on week 5 of training; that is, weeks after the improvement in general attentional performance (LDST and TMT-B scores obtained before the meditation sessions) was seen between weeks 1 and 3. One plausible explanation for the improvement seen with the early classes and sessions among both ReST and CMT participants is that some basic level of skill is necessary before beginners can apply the given practice instructions in a reliable and flexible way. A possibly complementary explanation is that the novelty of meditation in the first few weeks of training obscures any differences related to the specific practice. Perhaps just stopping demanding work and routines and listening to an instructor's voice was enough to stimulate restorative processes in those first weeks of training; that is, a bottom-up driven mindful state. The influence of contextual characteristics of the experience might thus be more important than the specific practice instructions in the very early stages of meditation training. After all, only once some basic level of meditation skill has been established can individuals be expected to regulate the meditative state top-down and thus begin to build a degree of independence from the setting. Only with top-down regulation of focused attention practice would signs of effort appear.

Given the lack of signs of effort with the classes and sessions in the first weeks of training, we suggest that the introduction of regular restoration with the in-class and homework meditation exercises rather than attention network training could have driven the generalized attentional improvements seen in those first weeks. Restoration could potentially also have influenced the generalized outcomes in other studies that have trained beginning meditators over very short time-spans (e.g., Friese et al., 2012; Tang et al., 2007; Zeidan et al., 2010).

In contrast with attention network training, Posner, Tang and colleagues (e.g., Posner et al., 2010; Tang and Posner, 2009) have proposed the term "attention state training" for those interventions that improve attention without targeting enhancements in attentional brain networks through repeated effortful activation. They include both low-effort open monitoring-type meditation and restorative nature experience in this category. We acknowledge the similarities between open monitoring and restorative nature experience but also see important differences between the two approaches, specifically in how they see the connection with present experience being supported by top-down (practice) and bottom-up (environmentally driven) processes. Different meditation practices can apparently hinder or facilitate restoration in the short-term, or train restoration skills over several weeks as we have shown with ReST. However, restoration can also take place independent of any specific practice and does not necessarily involve training in any skills. As seen from the environmental perspective, meditation can entail a needed change in setting and activity that supports psychological distance from demands and routines together with engagement with softly fascinating external or internal stimuli. In line with Kaplan (2001), we thus propose that restoration should be seen as a third distinct pathway to attentional benefits of meditation.

Restoration could be particularly prominent as a contributor to attentional improvements in the very early stages of meditation training and among individuals with a greater need for restoration; that is, those with more pronounced stress and concentration problems. These would be many of those individuals to whom mindfulness training is commonly recommended as a remedy. In contrast to CMT, the ReST approach seems to enhance the ability to reap restorative benefits during meditation, and improves attention over several weeks of training without imposing any potentially prohibitive attentional effort in the short-term. We address the acceptability of the two courses elsewhere (Lymeus et al., in preparation).

Regular nature experiences and ReST training can help ensure vulnerable individuals access to a fuller measure of their capabilities. Once sufficiently deep and frequent restoration has been ensured and the individual has enough resources to invest, attention network training could further enhance attention capabilities. With sufficient access to the needed resources, the effort of conventional mindfulness meditation will be more acceptable and can function as an important source of training and learning opportunities rather than as a perceived obstacle in meditation. One of the several potential uses for ReST could thus be as a preparation for conventional mindfulness training for particularly vulnerable individuals.

4.2. Future directions

In this paper we have only been able to speak to the contrast between ReST and our generic CMT course that emphasized focused attention exercise. Researchers might also consider contrasting ReST with a course that emphasizes open monitoring exercises without drawing on a restorative environment for support. However, we chose not to do so because (1) a strong emphasis on open monitoring meditation is not as common in conventional mindfulness courses for beginners, and (2) open monitoring is not as strongly connected with theory describing a clear pathway toward attentional enhancements among beginning meditators.

Our conclusions are based on performance changes on the LDST and TMT. We carefully selected these validated and widely accepted assays of selective and executive attention for their reliability, sensitivity, and efficiency. However, use of an extended or alternative test battery in replications could further inform the interpretation of the effects, and validate and extend our results.

We used nature-rich garden environments as meditation settings. In contrast, some traditional meditation gardens are quite sparse in visual elements and natural features. We do not consider richness as such to be the most relevant aspect of the environment, but rather the potential the environment holds to evoke a sense of being away and engage soft fascination. Perhaps a more austere garden is better suited to the needs of more experienced meditators who already have sufficient skills to direct and sustain attention to a more limited and subtle range of stimuli. Further, we do not suggest that garden environments or natural environments generally are the only kinds that can support meditation. In fact, we direct researchers to consider the many different ways that meditators, meditation teachers, and meditation institutions tend to contextualize the practice in space and time. Contextual factors could influence the meditation directly as impressions in one or more sensory modalities as well as through the personal, social, and cultural meanings that they convey.

Our replication of several important results from Study 1 with the improved design in Study 2 permits some confidence in the conclusions. However, confidence could be further reinforced with a controlled pre-meditation fatigue induction, a blinded design, additional control for motivation, and reliable assessments of potential mediating constructs in immediate connection with the meditation sessions. It remains for further research to determine to what extent the generalized attentional benefits of ReST are comparable to those seen with conventional mindfulness training also over longer time-spans or with individuals who do not experience any acute need for restoration. Further research is also needed to determine the degree to which the improvements seen with the two meditation approaches generalize also to other assays of relevant attentional functions and of emotion and stress, and whether the effects of the two meditation approaches translate similarly to improvements in the ability to manage real day-to-day demands.

5. Conclusions

The meditation setting can influence the processes and outcomes of meditation. The ReST approach to mindfulness training draws on restorative qualities in natural environments to support beginners with stress or concentration problems in effortless meditation. ReST seems to gradually enhance the ability to reap restorative benefits during meditation. Effortful focused attention meditation is thus not the only pathway of attentional improvements in early stages of meditation training. With focused attention meditation in CMT, signs of effort in meditation appeared gradually and only after early generalized improvements in the first weeks of training, against predictions based on the attention network training account. In practice, ReST holds promise as a relatively undemanding introduction to mindfulness training that may be particularly suitable for individuals with stress or concentration problems. ReST training could also be used to enhance restoration in limited contacts with nature.

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